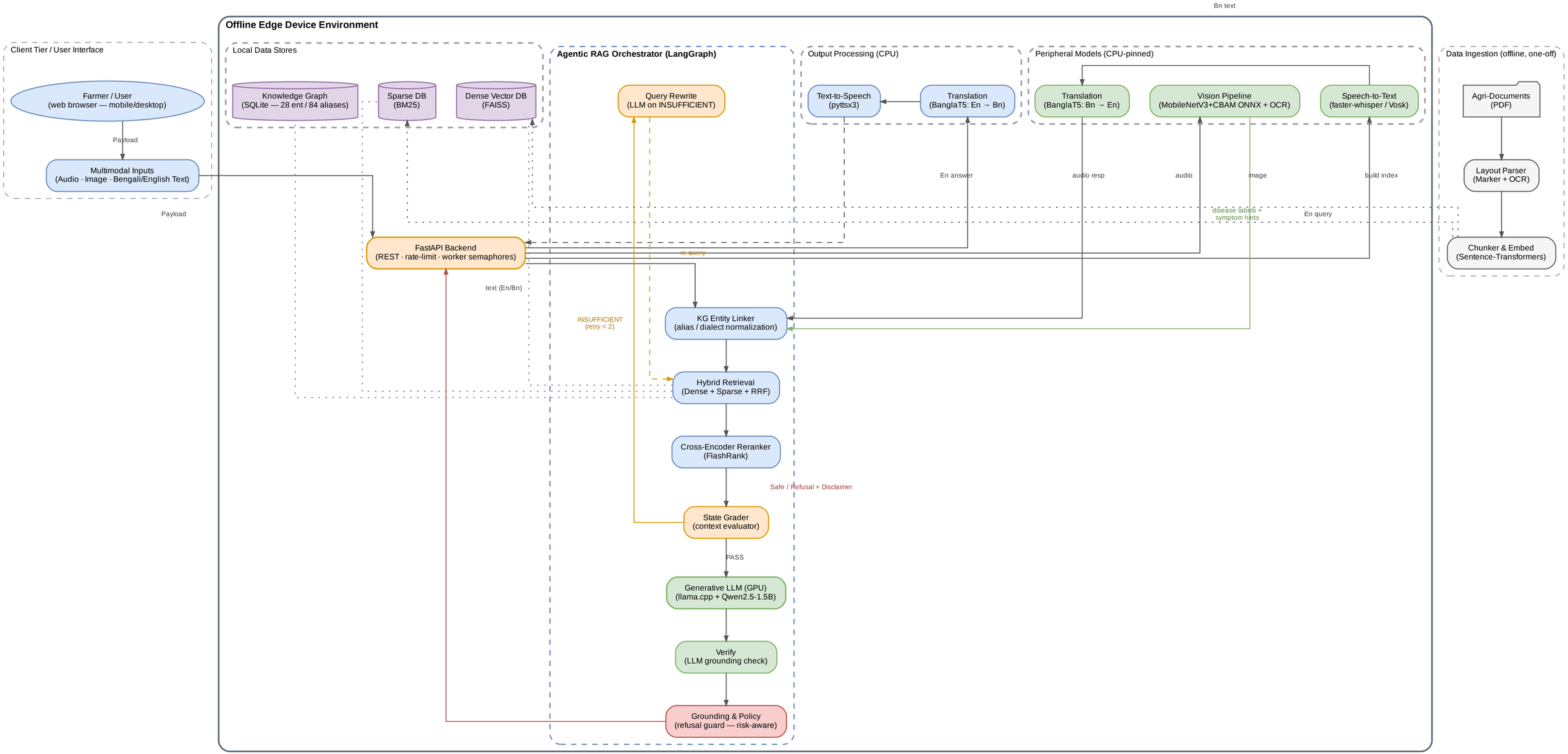
Offline Edge Device Environment
Client Tier / User Interface
Farmer / User
(web browser — mobile/desktop)
Payload
Multimodal Inputs
(Audio · Image · Bengali/English Text)
Payload
Local Data Stores
Knowledge Graph
(SQLite — 28 ent / 84 aliases)
Sparse DB
(BM25)
Dense Vector DB
(FAISS)
Agentic RAG Orchestrator (LangGraph)
Query Rewrite
(LLM on INSUFFICIENT)
Output Processing (CPU)
Text-to-Speech
(pyttsx3)
Translation
(BanglaT5: En → Bn)
Peripheral Models (CPU-pinned)
Translation
(BanglaT5: Bn → En)
Vision Pipeline
(MobileNetV3+CBAM ONNX + OCR)
Speech-to-Text
(faster-whisper / Vosk)
Data Ingestion (offline, one-off)
Agri-Documents
(PDF)
Layout Parser
(Marker + OCR)
Chunker & Embed
(Sentence-Transformers)
Bn text
En answer
audio resp
audio
image
build index
disease labels +
symptom hints
En query
FastAPI Backend
(REST · rate-limit · worker semaphores)
re-query
text (En/Bn)
INSUFFICIENT
(retry < 2)
KG Entity Linker
(alias / dialect normalization)
Hybrid Retrieval
(Dense + Sparse + RRF)
Cross-Encoder Reranker
(FlashRank)
Safe / Refusal + Disclaimer
State Grader
(context evaluator)
PASS
Generative LLM (GPU)
(llama.cpp + Qwen2.5-1.5B)
Verify
(LLM grounding check)
Grounding & Policy
(refusal guard — risk-aware)